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Promoting financial inclusion for savings groups: A financial education programme in rural Rwanda☆

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ABSTRACT

This study evaluates CARE Canada's financial education programme for members of savings groups in rural Rwanda two years after the intervention. The programme was designed to help group members gain better financial understanding and to encourage the usage of formal financial services. My difference-in-differences estimation only allows me to rule out an effect larger than 0.33 points of a financial literacy score measuring participants' ability to calculate financial trade-offs. This is large considering an average score of 1.7 at baseline. Among the final impacts, I find no evidence that the intervention impacts formal savings, assets, or subjective resilience, with the possible exception of a change from informal borrowing to formal borrowing. Overall, my findings suggest a mixed picture for usage of formal financial services.

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1. Introduction

Innovations in financial products, such as mobile money solutions and agent banking, now offer affordable and instant financial services, in particular for the rural poor in developing countries. Yet low financial literacy – or the ability to make informed decisions about financial resources – may limit people's ability to fully access these opportunities. In sub-Saharan Africa, only 43% of adults have an account with a bank or with a mobile money service provider, leaving nearly 340 million adults unbanked (Klapper et al., 2019). Financial literacy can help them to improve decision making regarding financial services in line with their true preferences and personal goals (FinScope, 2020). In response to the increasing relevance along with the low level of financial literacy, especially in developing countries (see Xu and Zia (2012) and Lusardi and Mitchell (2014), for reviews), CARE Canada partnered with the Mastercard Foundation to provide financial education and to facilitate access to financial institutions for young adults and women in Burundi, Côte d'Ivoire, Ethiopia, and Rwanda.

I evaluate CARE Canada's financial literacy training in rural Rwanda, where the programme was implemented by CARE Rwanda (henceforth CARE) and targeted 300,000 people between 2013 and 2017. It was designed for members of Village Savings and Loan Associations (VSLAs) who are not traditionally reached by formal financial means. My difference-in-differences (DiD) approach is based on a representative sample of Rwandan VSLAs. While more assumption-intense in terms of the narrow identification of the causal impacts, this impact evaluation benefits from its proximity to real-world policy and is a good approximation of what a scaled intervention can achieve.

The financial literacy training consists of seven sequential modules to equip participants with the skills to improve their money management but also to make them understand how formal financial markets work, and so to ultimately improve financial inclusion. It took place in the local language and was administered on two levels. First, CARE trained Village Agents (VAs) in the financial literacy content. VAs are local VSLA members, who usually get paid to facilitate and train VSLAs in their community and neighbouring areas. Second, VAs delivered the

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training to VSLA members for whom I collected data and measure impacts.

In Rwanda, while access to financial services is growing, understanding, uptake, and continued use of financial services remain a significant concern (FinScope, 2020). Indeed, 28% of adults do not borrow, and 61% rely on informal mechanisms, such as money-lenders who are often associated with extortive rates and non-transparent terms and conditions. Only 15% of adult Rwandans access credit through regulated formal lenders (FinScope, 2016).[2] Several studies have suggested a link between financial illiteracy and costly credit decisions (see Pak (2018), for a review) and between lifting credit constraints and agricultural productivity. Ali et al. (2014) show that many households in rural Rwanda are credit constrained and that lifting this constraint increases agricultural productivity by at least 17%.

This paper is most similar to the study by Sayinzoga et al. (2015), which also measures the impact of a standardised financial literacy training on financial behaviour in the southern province of rural Rwanda. The authors randomised access to the training for a non-random sample of village bank members. These members were already included in the formal financial sector,[3] and were chosen to be most suitable for participation by their peers or volunteered to participate. Their study, like mine, has a two-level training approach: their partner non-governmental organisation (NGO) trained village bank members, who were then explicitly asked by the NGO to share the information learned with their peer members. The authors refer to this second level as 'indirect training' or spillover effects. To probe the extent to which such spillovers exist, they select a random sample of village bank members and find no effects. In line with expectation and given the much more direct exposure to the treatment of the first level, they find strong effects on financial literacy, savings, borrowings, and new business start-up for village bank members on the first level. Using a two-stage regression approach, such that financial literacy is instrumented by the training dummy, the authors provide novel insights that enhanced financial literacy is one major channel of their findings.

In the present paper, I therefore complement Sayinzoga et al. (2015) by testing the effectiveness of a financial education intervention on the second level — the VSLA member, also referred to as 'spillover sample' by Sayinzoga et al. (2015). Both interventions' theory of change desire impacts not only to materialise for those who get trained by the implementing organisation but also for their peers. In other words, delivering training to only a small group of units will eventually be beneficial to many more in their environment. In my study, this means that the training stimulates responsible financial behaviour not only for the VA but also for VSLA members. For the intervention to be successful, better money management skills of VSLA members are the ultimate goal and, thus, the target population of my study. Moreover, I contribute evidence from a more representative sample of the rural poor in Rwanda than Sayinzoga et al. (2015). While their results are extremely important for the targeting of such trainings, they acknowledge that members on the first level "are different from their peers [...] and arguably they stand to gain more from the training than the average village bank member" (p. 22). Because these bank representatives tend to belong to the elite of the bank in terms of assets and education, at baseline, my sample is relatively less educated, less wealthy, and less financially included.

Early impact evaluations – mainly conducted in developed countries – showed little or no positive effects, especially for low-income samples and low-income study participants (see Fernandes et al. (2014) and Kaiser and Menkhoff (2017), for meta-analyses). Micro-based and cross-country studies in developing countries suggest a positive role of financial literacy for financial inclusion (Cole et al., 2011; Doi et al., 2014; Jamison et al., 2014; Grohmann et al., 2018). To account for the large increase of impact studies, especially in developing countries, a recent meta-analysis by Kaiser et al. (2021) solely draws inferences from 76 randomised experiments. 60.8% of all estimates reported are from programmes studied in developing economies. Kaiser et al. (2021) find an average impact of financial education programmes that is five times larger than the effect reported by Fernandes et al. (2014). In contrast to earlier meta-analyses, lower effects for low-income clients and a rapid decay of realised treatment effects cannot be confirmed.

I furthermore contribute to the growing literature on microcredits by examining the extent to which the effectiveness might be increased by complementary training measures. Recently, simple provision of microcredit has been debated in the literature. A set of randomised controlled trials (RCTs) published in the *American Economic Journal: Applied Economics* has aroused much attention and has been lauded (Banerjee et al., 2015) but also heavily criticised (Bédécarrats et al., 2020; Dahal and Fiala, 2020; Morduch, 2020). One line of argumentation suggests that the design of the microfinance programme matters a lot and the context in which it takes place as well as the type of customers that are targeted (see e.g Wydick (2016) and Bédécarrats et al. (2020)). The level of financial literacy and prior experience is likely to play an important role.

The paper proceeds as follows. Section 2 introduces the country background, the intervention, and the theory of change. Section 3 describes the data and the research design. Section 4 presents results. I discuss the main results and conclude in Section 5.

2. Country background and intervention

2.1. Country background

Southern Rwanda is a rural area where many people save in informal groups such as tontines or VSLAs. VSLAs are informal savings led microfinance groups. Every 10th person listed as an VSLA member in Africa lives in Rwanda (Care, 2017). Members meet once a week to purchase shares (save), borrow money, and repay loans. With no more than one or two hundred Rwandan Francs (RWF) per week, savings are small (less than 0.25 USD). Money is kept in a group lockbox. One savings cycle is a predefined period of about eight to twelve months. At the end of the savings circle, each member receives their share-out of the fund and their accumulated savings.[4]

Despite VSLAs being an important financial tool for the most vulnerable, the ultimate aim of policies for financial inclusion is increased access, understanding, uptake and continued use of formal financial products and services.

As early as 2008, the government of Rwanda initiated a long-term financial sector reform called the Financial Sector Development Programme, which aimed to "develop a stable and sound financial sector [...] to address the development needs of the economy and reduce poverty" (MINECOFIN, 2012, p. 26). This programme appears to have been remarkably successful, with FinScope (2012) revealing an increase in financial inclusion from

[2] In 2020, FinScope (2020) shows a slight increase of formal credit to 22%. Nevertheless, I rely on data from 2016 because the data better fit into this study's time horizon and context (FinScope, 2016).

[3] FinScope (2020, p. 25) defines Umurenge SACCOs, mobile money providers and insurance companies as non-bank formal financial institutions.

[4] For a detailed description of the VSLA model, see Karlan et al. (2017).

48% in 2008 to 72% in 2012, and FinScope (2016) showing further progress to 89% in 2016. Financial inclusion means uptake of financial products or mechanisms, including those who are banked (26%), those who take up other formal financial services such as mobile money, savings and credit cooperatives (SACCOs), and insurance companies (65%), and those who take up informal services (72%). The rise in financial inclusion is mainly driven by the proportion of adults who are now served by non-bank formal financial institutions. Nevertheless, the level of financial inclusion still varies significantly between districts, is lower in rural areas, among women, the less educated, poorer, and young people (FinScope, 2016). Building on the success of the first financial sector reform, the Rwandan government launched a second reform in 2012 with a particular emphasis on financial inclusion.

2.2. Intervention: Promoting financial education in Rwanda

Against this background, from 2013 to 2017, CARE Canada initiated a financial inclusion programme in Burundi, Côte d'Ivoire, Ethiopia, and Rwanda facilitating access to formal financial services for the poor in rural areas. In Rwanda, the programme was tailored to the needs of VSLA members in Ubudehe categories one and two. The Rwandan population is officially categorised into four different Ubudehe categories according to their wealth status, with the first designated for the poorest and the fourth for the wealthiest members of society.

I evaluate the effectiveness of the financial literacy training that was incorporated into the wider financial inclusion programme. CARE trains VAs in financial education to target VSLAs that had completed at least one savings cycle.[5] In Rwanda, VAs are local VSLA members (preferably a chairperson, secretary or a committed VSLA member) who facilitate VSLAs. VAs work on a fee for service basis. Established in 2009, the VA business model is designed to be cheap and to ensure quality service delivery as well as sustainability of VSLAs in rural areas. Each VSLA is regularly visited by the same (unique) VA. Each VA; however, serves multiple VSLAs in their community and neighbouring areas. As a result of their prior work to form, coach and supervise VSLAs, VAs have achieved a level of influence and recognition in the community. Their knowledge, exposure and understanding of VSLA methodology makes them a credible and trustworthy candidate to equip VSLAs with financial literacy skills. They know best what language to use and habits that comprise the appropriate mode of interactions with the training participants.

Table Appendix A.2 shows that the standardised training consists of seven sequential modules covering the following topics: why and where to save, how to set priorities for money and how to budget, how to make good choices about using financial institutions and to manage loans from banks or other non-bank financial services providers (FSPs). Each session takes place in the local language and lasts between one hour and a half to three hours and a half — that is more intense than comparable two-day trainings (see e.g., Calderone et al. (2018)). The intervention uses a classroom setting and a combination of methods, including group discussions, customised role plays, stories, and images.

The target group implies an explicit gender focus because 77% of VSLA members at baseline are women. Moreover, the training's manual is designed to make it especially easy for women to identify themselves with the themes and topics discussed during each session. Images and stories often document women in typical situations during their daily life; and thus make the training gender sensitive and appropriate for the VSLA context. Most trainers are women, providing women with positive female role models.

[5] VAs received a three-day training in the financial literacy content and financial linkages and were then trained to explicitly target VSLAs. Further, a detailed, 80-pages long training guide was provided to the trainer.

2.3. Theory of change

This financial education intervention can affect financial behaviours through various dimensions of financial literacy: financial attitude, financial awareness, and numeracy skills (Carpena et al., 2019). First, the training covers the advantages and disadvantages of different places to save or borrow, considering costs, ease of use, reliability, and availability of services. It discusses the inherent limitations of VSLAs to provide financial services such as lack of long-term savings options, safety of funds in the box, and small loan amounts for only a short duration. Realising the benefits of formal FSPs, including access to larger loan amounts and security of savings, may cause participants to change their attitude towards FSPs. Second, the training puts great emphasis on financial awareness, including budgeting and other financial planning skills. Third, the training covers the costs of borrowing (e.g., interest rates, fees, and time to apply). Even if interest rates are not directly calculated, the relevance and size of interest rates are prominently discussed; hence, strengthening participants' confidence to take out larger loans.

Overall, the three dimensions of financial literacy can enhance participants' ability to make better-informed financial decisions, to smooth their consumption, to invest in household assets or income-generating activities, and to ultimately increase their resilience against economic shocks (Drexler et al., 2014; Kaiser and Menkhoff, 2017; Berry et al., 2018; Calderone et al., 2018). Fig. 1 visualises the theory of change.

3. Survey data and empirical design

3.1. Data collection and variables

In early 2015, CARE provided a complete list of approximately 4000 active VSLAs in southern Rwanda. This list served as the sampling frame. Stratified by district and proportional to the number of VSLAs per district, 300 VSLAs were sampled. At the time when field work started, five of those 300 VSLAs were already inactive, six were unavailable and seven could not be contacted due to time and budget constraints, resulting in a total sample of 139 VSLAs in the control group and 144 in the treatment group. During May and June, the survey team interviewed a random sample of five members in each VSLA, for a total of 1405 individuals.[6] This was done by first compiling a list of all active members of the visited VSLA. Using smart mobile devices, a random number generator then selected ten names from this list. Out of ten, five acted as contingency in case the person could not be found or had already left the VSLA. Interviews were in-person with one VSLA member per household and using structured survey instruments. Each interview took about 45 min.

According to the global reach by Care (2017), VSLAs in Rwanda have on average 29 members of which 79% are women. At baseline, 77% are women and selected VSLAs have on average 28 members, which suggests a representative sample of Rwandan VSLAs.

A follow-up survey was implemented from May to June 2017. Figure Appendix B.1 shows the participation flow. The survey team was able to reach 241 VSLAs and 1174 VSLA members. 42 VSLA-level interviews could not be conducted (23 because VSLAs were no longer active), 212 respondents were unavailable (Figure Appendix B.2: 83% of them had either left the group or the VSLA was no longer active), and 3 respondents refused to participate in the survey. I see some shifting of individual members; some of them dropped out of the baseline sample, and others entered the

[6] In 95% of the cases, the intended number of members per VSLA could be reached. Overall, a minimum of four respondents were interviewed at all VSLAs.

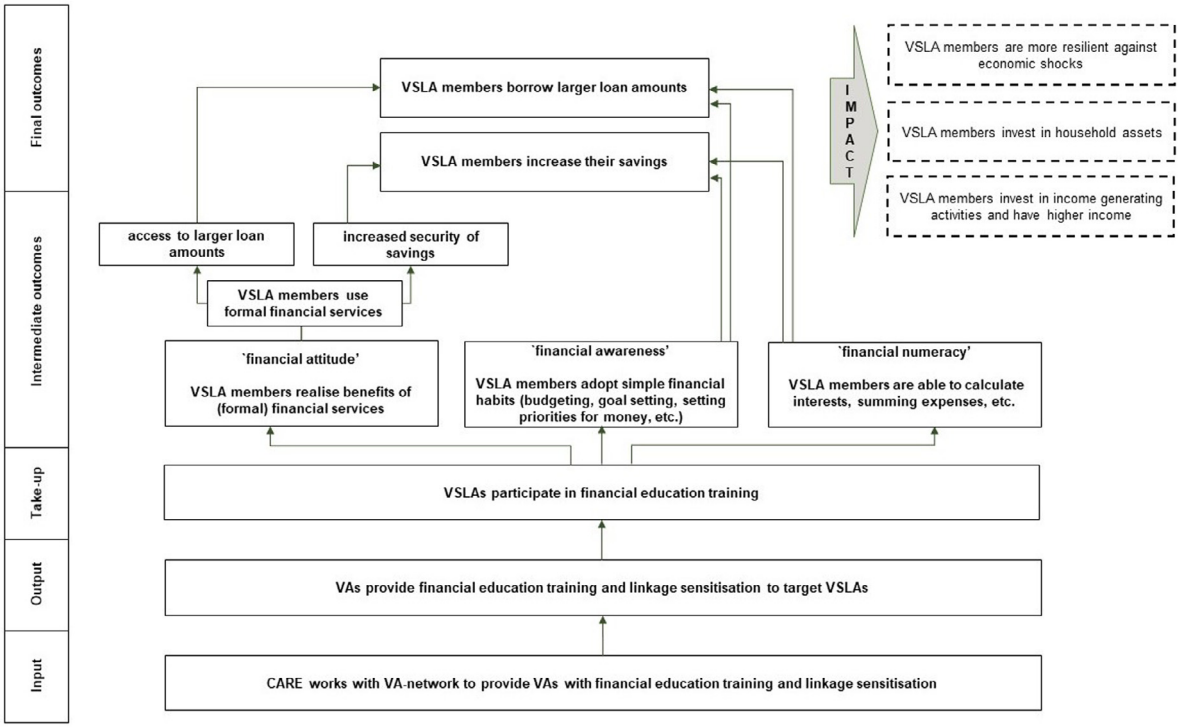

Fig. 1. Theory of change.

follow-up sample from the pre-defined contingency list. Out of 1174 completed interviews, 868 respondents were found at both base- and follow-up and 306 respondents were taken from the contingency list (henceforth replacements[7]).

I use the following primary outcome measures in my analysis: a financial literacy score, a binary outcome indicating whether someone has accumulated formal savings, and a formal borrowing-index (dummy for formal loans, number of formal loans, average formal loan amount).[8] I also examine three ultimate impact indicators (henceforth 'secondary outcomes'): monthly expenditures, an asset-index, and a resilience-index (ability to cope with emergencies, food security). For both the formal borrowing-index and the resilience-index, components are standardised, added and then standardised again. This double standardisation implies an equal weight to each component of the index. The asset-index is a z-score that is the first principal component of all common durable household assets. This approach for reducing and weighting the number of assets in a single wealth indicator has been recommended by Filmer and Pritchett (2001) and McKenzie (2005). In comparison with the double standardisation, assets that are more unequally distributed across the sample have a higher weight in the first principal component and yield a reasonable proxy for participants' wealth. My analysis was not pre-registered in a formal pre-analysis plan, but secondary outcomes are documented in an inception report that was completed before follow-up data collection and shared with the client of this evaluation (Table Appendix A.1). Table Appendix A.3 describes all outcomes.

The financial literacy score is an important intermediate result. It is based on an existing measure introduced by Lusardi and Mitchell (2011) and further developed by Cole et al. (2011). This approach focuses on numeracy skills for calculating financial trade-offs, comprising three questions on interest rates, inflation, and borrowings. As much of the training is however revolved around financial attitudes, it might be possible that dimensions of financial knowledge other than numeracy skills have changed (Anderson et al., 2017; Grohmann et al., 2022; Mangrum, 2021). Both, improved financial attitudes and awareness, can still have serious implications for VSLA members' decisions about savings, borrowings, and asset accumulation.

3.2. Identification strategy

Even though the initial assignment to treatment was random, the effective roll-out was determined by many practical factors ranging from logistical issues to monitoring difficulties. To probe into the impact of the financial literacy training on financial behaviour, I had to move to a non-randomised DiD approach that relies on the staggered order in which some VSLAs got trained first.

The DiD evaluation design requires at least two rounds of data collection. This implies that replacements, for whom I only observe endline data, are negligible. Nevertheless, I explore replaced people at endline for two reasons: (i) some outcomes were only collected at endline and including replacements in a cross-sectional analysis could increase statistical power, and (ii) comparing replacements with VSLA members who were baselined can inform about selective attrition as this has implications for the validity of my results. To account for the fact that the results may be biased due to attrition, I utilise inverse probability weighting (IPW) (Fitzgerald et al., 1998; Wooldridge, 2002). First, I estimate the probability of being found at endline using a binary logit model (including a range of covariates at baseline). Standard errors are clustered at the VSLA level. Second, I use the inverse of this predicted probability as a weight in the outcome analysis. Consequently, this impact evaluation features a mix of two different empirical methodologies: (i) a DiD approach using baseline

[7] According to enumerator comments at endline, 24% of baselined VSLA members were replaced because they had left the group. As enumerator comments are incomplete and because of constant fluctuations within VSLAs, I consider this to be a lower bound of the real VSLA drop-out rate.

[8] One way to account for multiple hypothesis testing is to aggregate variables into families of outcomes. I do this for borrowing but can only look at one binary outcome for savings because of data limitations at baseline.

and post-training data, which is my preferred specification, and (ii) a cross-sectional approach only looking at participants at endline, including those who were not baselined. More formally, I use the following regression model to isolate causal impacts in a DiD analysis:

$$Y_{it} = \alpha + \beta_1 T_i + \beta_2 Post_t + \beta_3 (T_i \times Post_t) + \gamma S_j + \delta X_{it} + \epsilon_{ijt} \quad (1)$$

where Y_{it} denotes the different outcomes of interest for individual, i, at time t, S_j is a vector of district strata dummies, X_{it} is a vector of covariates, and ϵ_{ijt} represents the error-term that captures individual and group cluster unobserved effects. $T_i \in \{0, 1\}$ indicates whether participants are treated ($T_i = 1$), and $Post_t \in \{0, 1\}$ indicates whether observations are made after the intervention ($Post_t = 1$). I further interact a linear time trend with the treatment dummy to control for time-invariant unobserved differences between groups. Standard errors are clustered at the VSLA level. β_3 is the DiD estimator.

The cross-sectional model can be estimated via the following regression equation:

$$Y_i = \alpha + \beta T_i + \gamma S_j + \epsilon_{ij} \quad (2)$$

where Y_i denotes the individual outcome variable, i, at endline, while other variables are the same as those defined in Eq. (1). The treatment effect is estimated by β.

During the phase of this evaluation, CARE moved to a new management information software that caused difficulties to match VSLA IDs between the baseline and endline. As a result, the treatment assignment could no longer be clearly identified. To overcome this issue, a local consulting firm was contracted to call each VA to establish when and what modules of the financial literacy component they had taught VSLAs. In comparison with administrative or self-reported data, the data from these calls are the most reliable source of information to identify treatment indicators for this impact evaluation (Table Appendix A.4). According to these phone calls, 139 VSLAs were trained, 9 VSLAs were informally trained, 78 VSLAs were not trained, 43 VSLAs could not be reached via phone or were unavailable, and 14 VSLAs were no longer active. The sample of 868 VSLA members found in both survey waves reduces to 699 for whom treatment information is available. Those who were only informally trained or could not be contacted are dropped from the analysis.

3.3. Sample description and tests of balance

Table 1 presents summary statistics for both, this study sample and the Sayinzoga et al. (2015)-sample. The study by Sayinzoga et al. (2015) is most comparable to this one and, hence, a comparison of both samples can partially explain the differences in results. Of the 868 individuals who were found in both rounds of data collection, 77% are female, 72% are married, 59% can spell a simple word in their local language, and only 9% have some form of secondary education. Respondents live in households with about five members. On average, respondents belong to the second Ubudehe category that means they have a dwelling of their own or can rent one; but still belong to the poorest in society.

The sample by Sayinzoga et al. (2015) looks significantly different than this study sample. 49% are female, 76% are married, and 16% have completed secondary education. While the majority in both samples owns livestock, village bank representatives are more likely to own a TV, radio, bicycle, or motorcycle than VSLA members. The average monthly spending reported by VSLA members is $33 and therefore less than $42 reported by village bank representatives. When restricting the number of financial literacy questions to three that are comparable between both studies (Table Appendix A.5), participants correctly respond to on average 1.6 to 1.7 questions. Inherited by the different selection criteria of both studies, village bank representatives are already more included into the formal financial system, with 31% save in bank, 45% save in MFIs, and 48% have a SACCO loan. Even if VSLAs can provide their members with practical financial experience that they can leverage to successfully engage with formal FSPs, with less or equal 5%, here, these statistics are negligible small. This stresses the necessity for targeted financial inclusion programmes in Rwanda.

The DiD approach provides unbiased treatment effect estimates when, in the absence of treatment, the average outcomes for the treated and comparison groups would have followed parallel paths over time. One way to tentatively underpin this assumption is to check balancing of means in the pre-treatment period. To test this, I rely on two different data sources, a VSLA/group- and a household-level survey, and present means of baseline variables for the comparison group as reported prior to treatment, alongside the difference vis-à-vis the treatment group in Tables 2 and 3. I also show p-values of the difference in treatment-comparison means. These tests are presented for both samples, the DiD (Table 3) and the cross-section (Table 5 in Section 3.4).

Overall, I find balance between the two groups in both samples for VSLA characteristics. For all but one variable the null hypothesis of a difference at baseline is clearly rejected (Table 2). In the DiD sample, only three (marital status and possession of health insurance) out of 35 variables collected for the individual VSLA member are statistically different between treatment and comparison groups at 10% level. Both tables imply very similar treatment and comparison samples and, hence, some validation for the parallel trend assumption.

Throughout this study, I further implicitly assume that the Stable Unit Treatment Value Assumption (SUTVA) holds. At baseline, the average distance to the nearest VSLA was about 1.6 km linear distance, whereby each village had on average one active VSLA. Spillover effects on untreated households (e.g., from higher awareness for formal financial products to an increased purchasing power) can, therefore, only cause problems subject to very large treatment effects.

3.4. Attrition and external validity

Table 4 shows response rates at endline. As a result of constant fluctuations, the overall attrition level of 37% is high; even higher than in other studies working in a comparable group context (e.g., Burke et al. (2019): 8,5%) or comparable country context (Sayinzoga et al. (2015): 18%). By contrast, there is no selective attrition based on treatment status in this study. To help understand attrition, I present summary statistics in Table 5 for those participants who were found, those who could not be found and those who replaced them at endline. Because the latter group has missing baseline information, I only compare characteristics that I assume to be time-invariant. The goal is to ensure that the three groups are similar in terms of average (pre-treatment) characteristics. This is tested formally through an F-test of joint orthogonality, which tests whether the observable characteristics are jointly unrelated to drop-out rates. For the treatment group, I cannot reject this null hypothesis validating that there is no selective attrition at endline. In contrast, the test is rejected for the comparison group. Unfound individuals are more likely to be male, single, and to run their own business. Similarly, the tests reject that the observable characteristics of the found people are those of the replacements in both groups. Individuals who are new to the endline sample are relatively more literate, wealthier according to their Ubudehe status, and more likely to be an independent farmer. I still present cross-sectional analyses including

Table 1
Pre-intervention descriptive statistics of DiD sample.

	Study sample			Sayinzoga et al. (2014) -sample		
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
Basic characteristics						
Female	868	0.77	0.42	341	0.49	0.50
Age	868	43.75	13.21	341	40.41	10.87
Secondary education and more	868	0.09	0.28	341	0.16	0.37
Spell	868	0.59	0.49			
Single	868	0.06	0.25	341	0.14	0.35
Widowed	868	0.16	0.36	341	0.08	0.27
Divorced	868	0.06	0.24	341	0.02	0.14
Married	868	0.72	0.45	341	0.76	0.43
Depression	868	0.34	0.47			
Subjective happiness	868	0.75	0.44	341	0.64	0.48
Household size	868	5.13	2.03	341	5.70	2.42
No. Children				341	3.60	2.50
Dependents	868	0.83	0.38			
Toddlers (0–5)	868	0.70	0.85			
Pupils (6–17)	868	1.84	1.63			
Pupils at school	868	1.72	1.55			
Ubudehe	868	2.00	0.54			
Health insurance	868	0.86	0.34			
Non-agricultural occ.	868	0.25	0.43			
Hh has a business	868	0.35	0.48			
Respondent has a business	868	0.33	0.47			
Assets						
Asset-Index	868	0.09	1.40			
Own land > 0.5 ha	868	0.44	0.50			
Land size in ha				265	1.25	1.90
Own livestock	868	0.89	0.31	328	0.86	0.35
Own house	868	0.90	0.30	336	0.90	0.29
Own TV	868	0.02	0.13	335	0.11	0.31
Own radio	868	0.62	0.49	339	0.91	0.28
Own mobile phone	868	0.48	0.50			
Own telephone				340	0.88	0.33
Own bicycle	868	0.19	0.40	337	0.41	0.49
Own motorcycle	868	0.01	0.10	337	0.08	0.27
Personal finances						
FL score	868	1.71	1.11	341	1.61	0.96
Has formal savings	868	0.27	0.44			
Save in Bank	868	0.02	0.15	339	0.31	0.46
Save in MFI	868	0.02	0.13	339	0.45	0.50
Any loan	868	0.98	0.15			
Formal borrowing-Index	868	−0.08	0.87			
SACCO loan	868	0.05	0.22	341	0.48	0.50
Monthly expenditures	868	31 441.95	43 808.29	341	40 435.85	40 715.62
Resilience-Index	865	0.15	0.89			

Notes: Values are calculated using baseline survey data collected for this study and in Sayinzoga et al. (2015).

replacements regardless, but with the aforementioned caveat that should be considered when interpreting these results.

Overall, the attrition rate is high; but note that the sample was neither screened for attrition before the intervention, nor are compositions of VSLAs static. Shifting of VSLA members is a function of several factors; most importantly, their ability to commit to weekly savings. As a result, 23 VSLAs dissolved within the two years of programme implementation. Additional individuals who decided to leave their group plus some usual attrition can yield an attrition rate that high. Table Appendix A.6 shows tests of balance looking at unfound people at endline, comparing those who left the VSLA and those who were unavailable for other reasons. There is balance across a wide range of measures, only employment characteristics show imbalance. Comparing those who were unavailable because they left the VSLA and those who were found at endline, I see that found VSLA members are more likely to live in greater households and to be slightly better off in terms of their Ubudehe category (Table Appendix A.7). This suggests that I cannot rule out any upward bias of the results.

Moreover, this evaluation benefits from its proximity to real-world policy because it was embedded into a business-as-usual setup. The probability of distortions by Hawthorne effects or special care problems as described by Peters et al. (2018) is thus very limited.

4. Empirical results

4.1. Primary and secondary outcomes

My primary interest lies in impacts on a financial literacy score, a formal savings indicator, and a formal borrowing-index. Despite grouping outcome measures into indices, in each table, I also report the familywise error rate adjusted p-values using the Westfall–Young stepdown resampling method (Westfall and Young, 1993). Table 6 presents regression results from Eq. (1) and Table Appendix A.8 presents mean comparison of the cross-sectional sample to conduct a comparison with Sayinzoga et al. (2015) (see Eq. (2)).[9]

Sayinzoga et al. (2015) detect strong positive impacts on financial literacy for village bank representatives but cannot validate

[9] In Tables Appendix C.1 through Appendix C.5, I show that results are robust to the inclusion of covariates unbalanced at baseline, to the imputation of missing values, and to the cross-sectional approach.

Table 2
Baseline characteristics and balance for group variables.

	Treatment			Control			Regression difference	p-Value
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.		
Panel (a) Difference-in-differences sample								
Active members	103	28.40	3.82	43	28.53	2.56	−0.310	0.650
Default frequency	103	1.70	0.81	43	1.58	0.76	0.158	0.316
Bank account	103	0.01	0.10	43	0.02	0.15	−0.007	0.798
SACCO account	103	0.78	0.42	43	0.81	0.39	0.007	0.926
No account	103	0.19	0.40	43	0.16	0.37	−0.026	0.720
Saved in Bank	103	0.03	0.17	43	0.09	0.29	−0.037	0.451
Saved in SACCO	103	0.73	0.45	43	0.77	0.43	0.011	0.902
Saved in MFI	103	0.07	0.25	43	0.05	0.21	0.035	0.392
Borrowed from SACCO	103	0.09	0.28	43	0.02	0.15	0.066	0.070
Panel (b) Cross-section with replacements								
Active members	123	28.37	3.69	68	28.41	2.71	−0.165	0.760
Default frequency	123	1.68	0.80	68	1.65	0.81	0.054	0.695
Bank account	123	0.01	0.09	68	0.03	0.17	−0.016	0.473
SACCO account	123	0.80	0.40	68	0.82	0.38	0.040	0.511
No account	123	0.18	0.38	68	0.15	0.36	−0.042	0.462
Saved in Bank	123	0.04	0.18	68	0.11	0.31	−0.052	0.209
Saved in SACCO	123	0.75	0.44	68	0.78	0.42	0.037	0.598
Saved in MFI	123	0.06	0.23	68	0.04	0.21	0.005	0.896
Borrowed from SACCO	123	0.10	0.30	68	0.01	0.12	0.081	0.010

Notes: Values are calculated using baseline survey data. The last column reports the *p*-value of the OLS regression of the listed baseline characteristics on the indicator for treatment plus district fixed effects.

these results for indirectly trained members in the same villages ('spillover sample'). My results are comparable to their spillover analysis. The effect on a financial literacy score is positive but not statistically different from zero. First, this may result from the fact that the score focuses on numeracy skills calculating financial trade-offs (Lusardi and Mitchell, 2011), and that the training content did not explicitly address numeracy skills. Second, the standard error on the score is high. Given the reached sample size, this insignificant result can also stem from imprecision of inference.

A second key outcome is whether the financial literacy training changed participants' savings behaviour. Although the formal savings coefficient is positive, it is not significantly different from zero. One reason may be that VSLA members make already rational contribution to savings given their income; reaching their saturation point with no scope for significant changes.[10] Alternatively, VSLAs already provide access to a safe and convenient place to save; hence, for participants the incentive to save with a formal FSP instead of a savings group is low. If this is true, the training should have pursued a different goal.

Another objective is to increase understanding of borrowing options, thereby facilitating access to larger loan amounts. I find an increase in the formal borrowing-index of 0.19 SDs. The effect is likely a combination of all the components of this index, including higher uptake of formal loans, an increase in the number of formal loans and average loan amounts. It becomes even more intense because the comparison group slightly lessens its formal borrowing behaviour (see Fig. 2). When correcting for sample attrition (see Section 3), the positive finding is less significant and slightly smaller with 0.16 SDs; when correcting for multiple hypothesis testing, it vanishes completely. The *p*-value quadruples from 0.03 to 0.14, which is quite a conservative adjustment but signals caution when interpreting this result. Further explorative analysis in Table Appendix A.9 breaks out the results of this index and looks at informal borrowings (columns four to six). The negative result in informal borrowings suggests a positive change in borrowings from informal to formal sources, including SACCOs, microfinance institutions (MFIs), mobile network operators (MNOs), and banks. Also, total borrowings appear to increase as the reduction in informal loan amounts is smaller relative to the increase in formal loan amounts. This net increase is based on noisy coefficients, but, more optimistically, the confidence interval also includes economically meaningful effect sizes. This suggests access to larger loan amounts as it was envisioned by CARE.

Table 7 presents findings for secondary outcomes, including monthly expenditures, a resilience-index, and an asset-index. The average monthly spending at baseline was about $33. At endline, this number increased substantially to about $41 for the comparison group but remained constant for the treatment group (see Fig. 3). The negative coefficient of about $10.6 could imply negative unintended consequences for participants from this intervention. It could also be a direct conclusion from increased overall borrowings. To report on the relevant mediator that may trigger these results, however, is beyond the scope of this study; so, this is purely speculative. Moreover, there is no evidence that the training had significant impacts on participants' asset-index or resilience-index.

4.2. Results comparison with Sayinzoga et al. (2015)

In Table Appendix A.8, I present additional results on savings outcomes to improve comparability with Sayinzoga et al. (2015). As detailed information on savings was only collected at endline, I can only compare my results to theirs when analysing the cross-section at endline — an analysis that needs to be interpreted with caution (see Section 3.4). A comparison of savings rates at endline yields similar results between the two studies, even more similar between the first level in Sayinzoga et al. (2015) (village bank representatives) and the second level in my study (VSLA members). This measure is the ratio between annual total savings and annual expenditures, multiplied by 100.

Moreover, both studies find increased formal credit. The binary variable for the uptake of any loan from the local cooperative

[10] As part of this impact evaluation, a series of focus group discussion yield the same result. Only one interview respondent did not report having trouble saving, with the rest suggesting that the lack of a consistent income prevents them from saving more.

Table 3
Baseline characteristics and balance for individual variables of DiD sample.

	Treatment		Control		Regression difference	p-Value
	Mean	Std. Dev.	Mean	Std. Dev.		
Basic characteristics						
Female	0.78	0.42	0.80	0.40	−0.025	0.533
Age	44.17	12.79	43.28	13.83	0.261	0.850
Secondary education and more	0.08	0.26	0.11	0.30	−0.023	0.336
Spell	0.59	0.49	0.60	0.49	0.037	0.461
Single	0.05	0.22	0.08	0.27	−0.032	0.134
Widowed	0.13	0.34	0.21	0.41	−0.051	0.090
Divorced	0.05	0.23	0.07	0.26	0.007	0.778
Married	0.76	0.43	0.64	0.48	0.076	0.095
Depression	0.34	0.47	0.34	0.47	−0.010	0.846
Subjective happiness	0.74	0.44	0.76	0.43	−0.022	0.587
Household size	5.15	2.00	4.93	2.13	−0.004	0.982
Dependents	0.84	0.37	0.79	0.41	0.014	0.686
Toddlers (0–5)	0.71	0.85	0.65	0.84	0.060	0.358
Pupils (6–17)	1.91	1.62	1.64	1.62	0.105	0.445
Pupils at school	1.73	1.48	1.60	1.57	−0.012	0.925
Ubudehe	2.02	0.54	2.02	0.52	−0.039	0.454
Health insurance	0.84	0.36	0.91	0.27	−0.059	0.072
Non-agricultural occ.	0.22	0.41	0.31	0.46	−0.083	0.127
Hh has a business	0.35	0.48	0.30	0.46	0.043	0.445
Respondent has a business	0.33	0.47	0.30	0.46	0.029	0.607
Assets						
Asset-Index	0.02	1.37	0.18	1.42	−0.155	0.285
Own land > 0.5 ha	0.44	0.50	0.45	0.50	−0.055	0.352
Own livestock	0.89	0.31	0.87	0.33	−0.016	0.612
Own house	0.90	0.30	0.91	0.28	−0.032	0.242
Own TV	0.01	0.12	0.03	0.18	−0.009	0.461
Own radio	0.62	0.49	0.64	0.48	0.004	0.925
Own mobile phone	0.45	0.50	0.52	0.50	−0.007	0.883
Own bicycle	0.16	0.37	0.22	0.42	−0.038	0.357
Own motorcycle	0.01	0.09	0.01	0.10	−0.002	0.796
Personal finances						
FL score	1.70	1.12	1.73	1.10	−0.070	0.575
Has formal savings	0.27	0.44	0.29	0.45	0.003	0.944
Save in Bank	0.02	0.14	0.04	0.19	−0.021	0.240
Save in MFI	0.02	0.14	0.03	0.16	−0.000	0.994
Any loan	0.98	0.15	0.98	0.15	0.003	0.830
Formal borrowing-Index	−0.13	0.79	−0.01	0.97	−0.115	0.157
SACCO loan	0.04	0.20	0.05	0.21	−0.008	0.646
Monthly expenditures	31 715.81	44 820.39	29 531.49	42 514.65	4062.179	0.314
Resilience-Index	0.14	0.86	0.23	0.91	−0.073	0.382

Notes: Values are calculated using baseline survey data. The last column reports the *p*-value of the OLS regression of the listed baseline characteristics on the indicator for treatment plus district fixed effects, with robust standard errors clustered at the VSLA level.

Table 4
Survey response rates.

	All found mean	Control found mean	Treatment found mean	Regression difference	p-Value
Assignment	0.63	0.62	0.63	0.007	0.854
Treatment	0.63	0.57	0.70	0.045	0.322

Notes: The first line refers to treatment assignment at baseline. The second line refers to treatment according to Village agent ('VA') phone calls — my preferred treatment variable. The last column reports the *p*-value of the OLS regression of either an indicator for assignment (1st line) or an indicator for treatment according to VA phone calls (2nd line) on the indicator for whether the individual was found at endline plus district fixed effects, with robust standard errors clustered at the VSLA level.

in Sayinzoga et al. (2015) is best comparable to the binary variable for the uptake of any formal loan in column one of Table Appendix A.9. Keeping in mind that individuals in Sayinzoga et al. (2015) are already linked to local cooperatives, the magnitude of the coefficient in their study is greater than in mine. If I compare my results to the spillover effects in Sayinzoga et al. (2015), that is looking at the effects of fellow village bank members who are drawn from a village bank where one other member has received training (and was told to share their knowledge), the magnitudes of both studies are becoming more similar. Whereas Sayinzoga et al. (2015) find modest impacts between 0.06 to 0.08 percentage points for village bank members, I find an impact of 0.05 percentage points for VSLA members.

4.3. Gender differences

Table 8 divides the panel into female and male, respectively. Like in the pooled sample, I neither see an effect on the financial literacy score nor on the formal savings dummy. For the borrowing-index, I find that this effect is driven by the impacts on both genders, though it is significant and greater for men. When applying attrition correction weights, statistical significance vanishes also for men. As for secondary outcomes, the coefficient on monthly expenditures is negative for female participants and robust when correcting for sample attrition. Again, this result stems from an increase in expenditures for women in the comparison

Table 5
Baseline characteristics and balance for individual variables of cross-section.

	Treatment			Control		
	Unfound mean	Found mean	Replaced mean	Unfound mean	Found mean	Replaced mean
Basic characteristics						
Female	0.82	0.78	0.82	0.69	0.80	0.74
Age	43.12	44.17	42.88	41.11	43.28	42.25
Secondary education and more	0.07	0.08	0.05	0.07	0.11	0.05
Spell	0.57	0.59	0.87	0.63	0.60	0.94
Single	0.11	0.05	0.04	0.17	0.08	0.06
Widowed	0.19	0.13	0.20	0.17	0.21	0.20
Divorced	0.06	0.05	0.03	0.04	0.07	0.06
Married	0.63	0.76	0.73	0.62	0.64	0.68
Depression	0.33	0.34		0.38	0.34	
Subjective happiness	0.82	0.74	0.87	0.79	0.76	0.91
Household size	4.53	5.15	5.02	4.74	4.93	5.18
Dependents	0.75	0.84	0.85	0.79	0.79	0.77
Toddlers (0–5)	0.58	0.71	0.70	0.74	0.65	0.48
Pupils (6–17)	1.43	1.91	1.74	1.51	1.64	1.84
Pupils at school	1.33	1.73	1.78	1.47	1.60	1.77
Ubudehe	1.99	2.02	2.45	1.90	2.02	2.47
Health insurance	0.84	0.84		0.87	0.91	
Non-agricultural occ.	0.23	0.22	0.05	0.27	0.31	0.10
Hh has business	0.31	0.35	0.27	0.38	0.30	0.35
Respondent has business	0.28	0.33	0.22	0.37	0.30	0.32
Sample size	201	479	124	164	220	114
Joint orthogonality test						
Unfound versus Found	0.180			0.006		
Found versus Replaced		0.000			0.000	

Notes: Tests for comparing found, unfound, and replaced study participants on district strata, with robust standard errors clustered at the VSLA level. 'Replaced' refers to a VSLA member who enters the sample at follow-up data collection because the person that was interviewed at baseline was not available or left the VSLA.

Table 6
Primary outcomes — DiD sample.

	(1) FL score (3) β/SE	(2) Has formal savings β/SE	(3) Formal borrowing-Index β/SE
DiD	0.043	0.015	0.190**
	(0.147)	(0.042)	(0.090)
	[−0.247	[−0.069	[0.013
	0.332]	0.098]	0.367]
DiD (IPW)[a]	0.030	0.020	0.164*
	(0.150)	(0.043)	(0.091)
	[−0.265	[−0.065	[−0.016
	0.325]	0.105]	0.343]
Observations	1398	1398	1398
Control mean	1.82	0.19	−0.07
Control SD	0.97	0.39	0.77
R-squared	0.009	0.021	0.015
P-WYoung	0.960	0.960	0.140

Notes: The table shows point estimates with robust standard errors clustered at the VSLA level (in parentheses) as well as lower and upper ends of 95 percent confidence intervals in brackets. Regressions include district strata. I impute missing values by setting them equal to the mean of the respective outcome variable for the relevant treatment group. Monetary values are winsorized at the 99 percentile.

[a]Presents results from the weighted least squares — correcting for sample attrition.

Table 7
Secondary outcomes — DiD sample.

	(1) Monthly expenditures β/SE	(2) Asset-Index β/SE	(3) Resilience-Index β/SE
DiD	−10 204**	0.037	−0.058
	(4118)	(0.128)	(0.111)
	[−18 326	[−0.216	[−0.276
	−2083]	0.290]	0.160]
DiD (IPW)[a]	−7952*	0.028	−0.027
	(4746)	(0.129)	(0.110)
	[−17 311	[−0.227	[−0.245
	1407]	0.282]	0.190]
Observations	1398	1398	1398
Control mean	39 997	0.05	0.06
Control SD	36 640	1.45	0.98
R-squared	0.018	0.005	0.024
P-WYoung	0.020	0.880	0.880

Notes: The table shows point estimates with robust standard errors clustered at the VSLA level (in parentheses) as well as lower and upper ends of 95 percent confidence intervals in brackets. Regressions include district strata. I impute missing values by setting them equal to the mean of the respective outcome variable for the relevant treatment group. Monetary values are winsorized at the 99 percentile.

[a]Presents results from the weighted least squares — correcting for sample attrition.

group and constant expenditures for women in the treatment group.

4.4. Sensitivity of results to statistical power

Early in the planning stages, I derived the sample size that is needed to achieve a certain level of statistical power. I drew on previous work on energy access in Rwanda by Grimm et al. (2017) and Lenz et al. (2017) to calculate the mean, standard deviation, and intra-class correlation (ICC) for indicators in line with the inception report: assets owned by the household, health and education expenses, and decision making at home. The ICC is important because members sampled from the same VSLA tend to share similar outcomes by the virtue of living in the same area. Less variability of responses within VSLAs artificially increases differences in outcomes between VSLAs, thereby reducing the statistical power. ICCs calculated here are low for some indicators (members within VSLAs are no more like each other than members from different VSLAs) and well above 0.1 for others (members within the same VSLA have similar outcomes). That

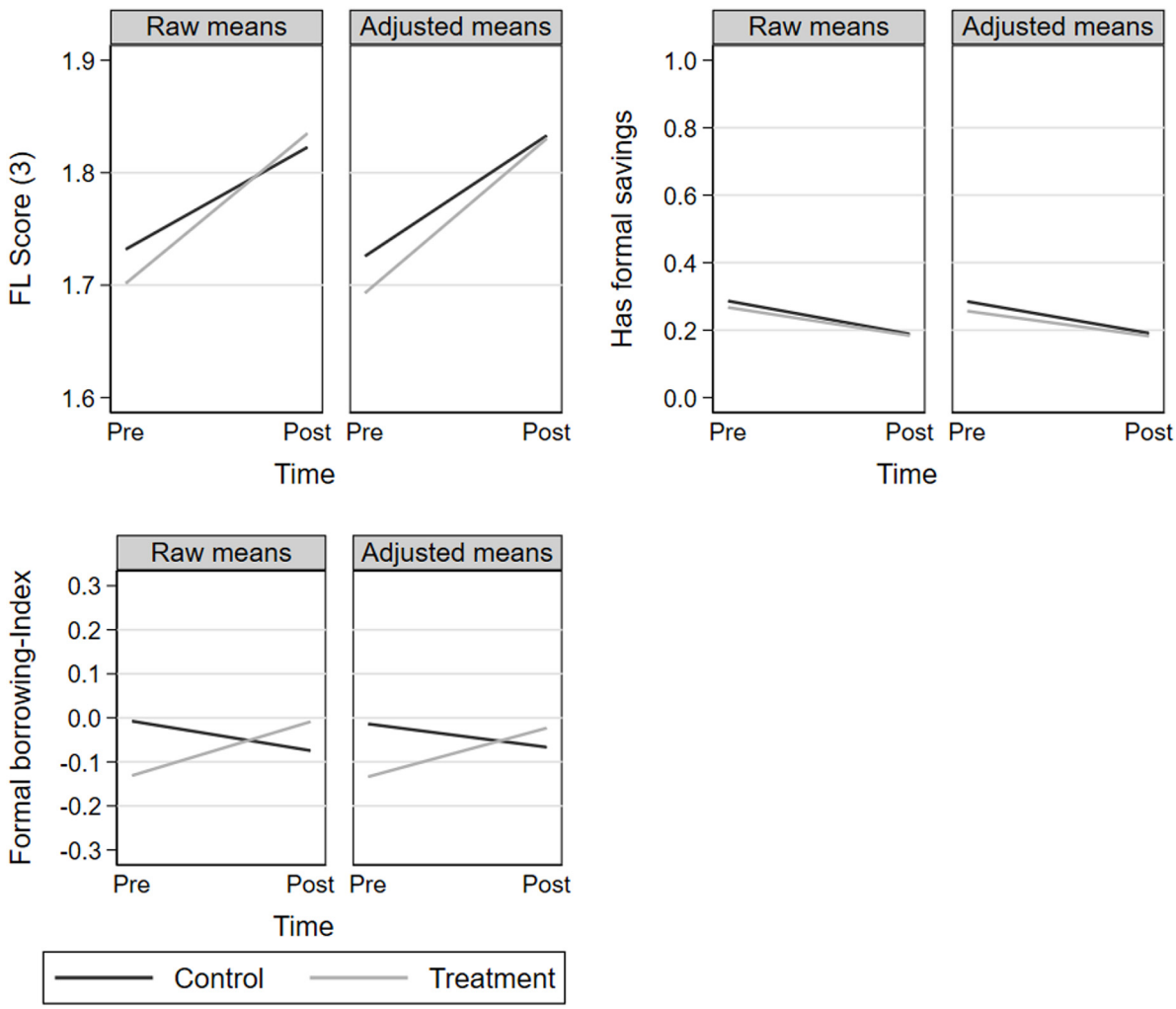

Fig. 2. Pre/Post graphs for primary outcomes.

Table 8
Gender differences — DiD sample.

	Primary outcomes			Secondary outcomes		
	(1) FL score (3)	(2) Has formal savings	(3) Formal borrowing -Index	(4) Monthly expenditures	(5) Asset-Index	(6) Resilience-Index
	β/SE	β/SE	β/SE	β/SE	β/SE	β/SE
Panel (a): Women only						
DiD	0.106	0.026	0.136	−9963**	0.022	−0.006
	(0.154)	(0.046)	(0.092)	(3902)	(0.149)	(0.125)
DiD (IPW)[a]	0.115	0.026	0.127	−7786*	−0.017	0.019
	(0.157)	(0.047)	(0.088)	(4404)	(0.148)	(0.124)
Observations	1092	1092	1092	1092	1092	1092
Control mean	1.77	0.18	−0.09	36 722	−0.03	−0.01
Control SD	0.98	0.38	0.80	34 290	1.49	1.01
R-squared	0.013	0.014	0.015	0.017	0.006	0.024
Panel (b): Men only						
DiD	−0.186	−0.019	0.390*	−12 380	0.056	−0.272
	(0.299)	(0.111)	(0.232)	(12 883)	(0.230)	(0.203)
DiD (IPW)[a]	−0.283	0.000	0.293	−9410	0.157	−0.208
	(0.306)	(0.104)	(0.253)	(15 163)	(0.246)	(0.219
Observations	306	306	306	306	306	306
Control mean	2.05	0.24	−0.00	53 479	0.36	0.34
Control SD	0.95	0.43	0.65	42 913	1.26	0.83
R-squared	0.034	0.062	0.043	0.032	0.024	0.042

Notes: Regressions include district strata, with robust standard errors clustered at the VSLA level. I impute missing values by setting them equal to the mean of the respective outcome variable for the relevant treatment group. Monetary values are winsorized at the 99 percentile.
[a]Presents results from the weighted least squares — correcting for sample attrition.

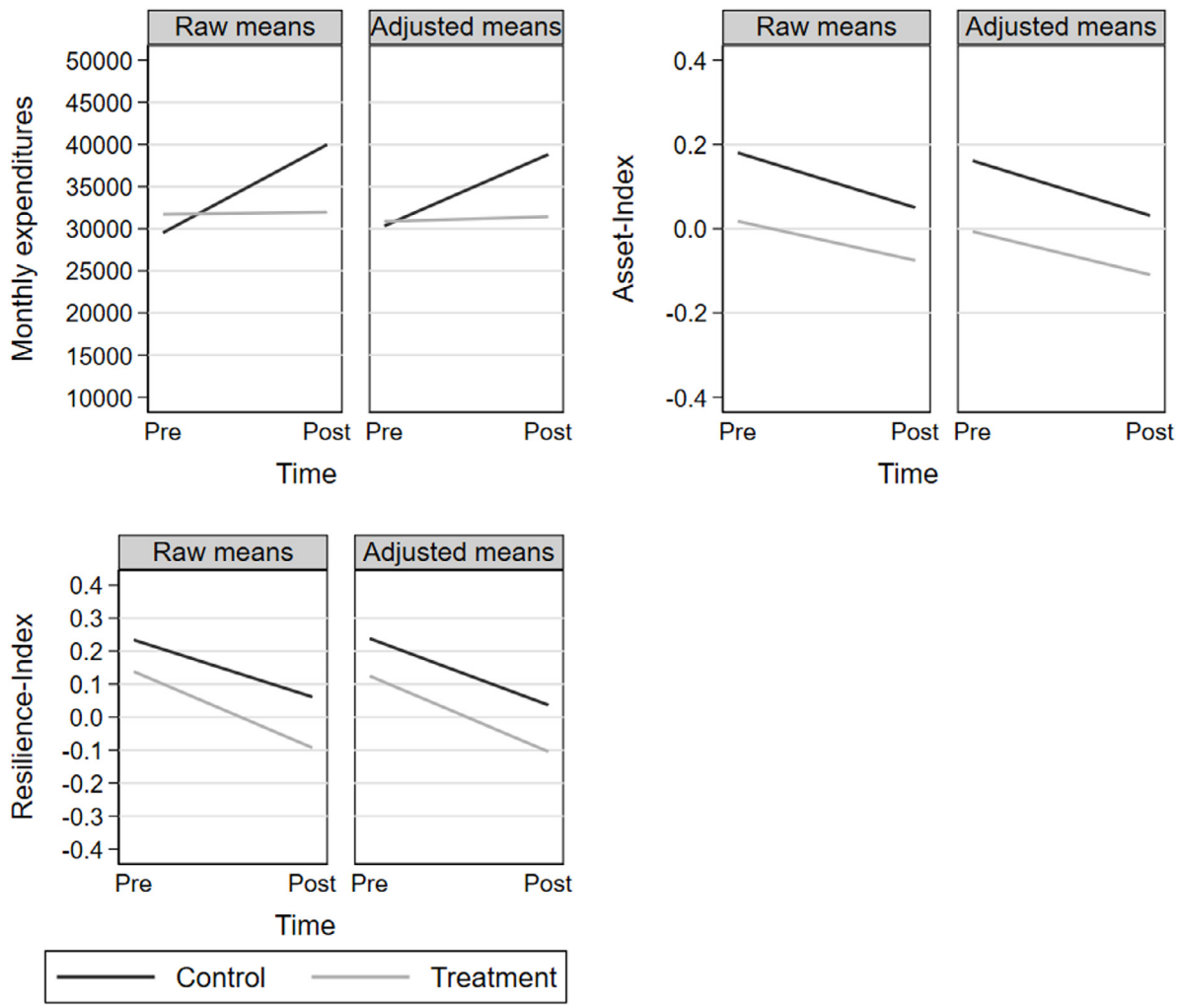

Fig. 3. Pre/Post graphs for secondary outcomes.

is why I look at two scenarios when visualising ex-ante power calculations: one scenario with an average of high ICCs (=0.13) and another scenario with an average of low ICCs (=0.05). In Figure Appendix B.3, the power analysis suggests that under usual assumptions a sample of 300 clusters (150 treatment, 150 comparison) with a random sample of eight individuals each is well powered to detect effect sizes of 15% in the conservative, high ICC scenario. At baseline, however, the sample size had to be reduced to 300 VSLAs with five members each.

At endline, the survey team reached a total of 241 VSLAs with 1174 member interviews. Of the 1174 interviewed, 868 participants were found in both survey waves. Figure Appendix B.4 shows the Minimum Detectable Effect (MDE) size for the binary indicator of whether someone has accumulated formal savings, given the realised sample size at endline and the calculated ICC at baseline. Accordingly, this study would only be able to detect a 21% increase in formal savings. In Table 6, the DiD estimator is insignificant at 0.015 (0.042). Given the actual, small sample size, I, therefore, cannot infer that there is no treatment effect in formal savings.

5. Discussion and conclusion

I analyse the efficacy of an intensive financial literacy and linkages training in Rwanda that covers simple economic and financial concepts. Because of deviations from initial experimental protocols, I am using a quasi-experimental approach to estimate causal impacts on three primary outcomes, including a financial literacy score, a formal savings indicator, and a formal borrowing-index. The study was embedded into a business-as-usual setup with all the associated challenges. First, CARE changed its identification software during the evaluation phase that created difficulties to retrospectively match VSLAs to their actual treatment status. To solve this, I rely on a mobile phone survey with the trainers (VAs) to clearly identify treated VSLAs and non-treated VSLAs. Second, the non-static compositions of VSLAs partially led to a relatively higher attrition rate than in comparable studies (Burke et al., 2019; Sayinzoga et al., 2015). I, therefore, apply attrition correction weights to all regressions.

Impacts of the financial literacy training are ambiguous on the financial behaviour of VSLA members. Unlike Sayinzoga et al. (2015), the financial literacy training seems to play a very limited role in equipping participants with the ability to calculate financial trade-offs. The effect on a financial literacy score is positive, but small and statistically insignificant. As much of the training is revolved around financial attitudes, it is likely that dimensions of financial knowledge other than numeracy skills have affected financial behaviour and remain undetected. The financial literacy training may improve formal borrowings; however, multiple hypothesis corrections reveal no effect. I also cannot determine strong effects on formal savings, assets, or subjective resilience. Monthly treatment expenditures are flat, while expenditures among the comparison group increased.

I conclude by pointing out why not all expectations of the implementer can be met by the data; and not all of it necessarily corresponds with an unsuccessful intervention. First, the small sample size at endline causes a low statistical power of this study. As a result, I cannot infer that there is no treatment effect from this intervention. Second, the financial literacy training did not affect my measure on financial literacy. On the one hand, this could result from the fact that the measure solely focuses on numeracy skills - a very common measure in the financial literacy literature (Lusardi and Mitchell, 2011). Because the training did not explicitly address these skills, it is unlikely to observe movement on this coefficient. However, calculating financial interests is a critical skill when requesting access to larger loan amounts. Like Carpena et al. (2019), this finding suggests that combining financial literacy education with mathematics training may be necessary to improve a person's ability to make financial calculations. On the other hand, the training approach might have been inadequate to address personal limitations for behavioural changes or lack of knowledge (Carpena et al., 2019). Implemented in group settings with a one-size-fits-all approach and no personalised feedback, the training may indeed be limited in its success to improve participants' financial literacy (Grohmann et al., 2022). Moreover, the amplitude of the training effects heavily depends on the quality and engagement of the trainer. Given limited financial resources, training VAs is likely the most cost-effective option; though, perhaps to the detriment of achieving optimal results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbef.2022.100662.

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